

hw2

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```
data <- read.csv("./datasets/Dataset.csv")
```

```
class(data['Energy_Consumption_kWh'])  
head(data['Energy_Consumption_kWh'])
```

```
null_model <- lm(Energy_Consumption_kWh ~ 1, data = data)
```

```
null_model$coefficients
```

```
## (Intercept)  
##      102.8555
```

```
mean(data$Energy_Consumption_kWh)
```

```
## [1] 102.8555
```

```
sse_null <- sum(resid(null_model)^2)  
sse_null
```

```
## [1] 16436.42
```

```
summary(null_model)
```

```
##  
## Call:  
## lm(formula = Energy_Consumption_kWh ~ 1, data = data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -28.178 -14.780   0.021  11.335  39.273   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   102.86       2.59   39.71  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 18.31 on 49 degrees of freedom
```

```
sim_lin_reg = lm(data$Energy_Consumption_kWh ~ data$Temperature_C, data = data)
```

```
summary(sim_lin_reg)
```

```
##  
## Call:  
## lm(formula = data$Energy_Consumption_kWh ~ data$Temperature_C,  
##     data = data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -12.5034  -2.4541   0.0241   3.4235   8.3998   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    50.93031     2.04439   24.91  <2e-16 ***  
## data$Temperature_C  2.45531     0.09158   26.81  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 4.63 on 48 degrees of freedom  
## Multiple R-squared:  0.9374, Adjusted R-squared:  0.9361   
## F-statistic: 718.8 on 1 and 48 DF,  p-value: < 2.2e-16
```

```
sse_lin <- sum(resid(sim_lin_reg)^2)
```

```
sse_null
```

```
## [1] 16436.42
```

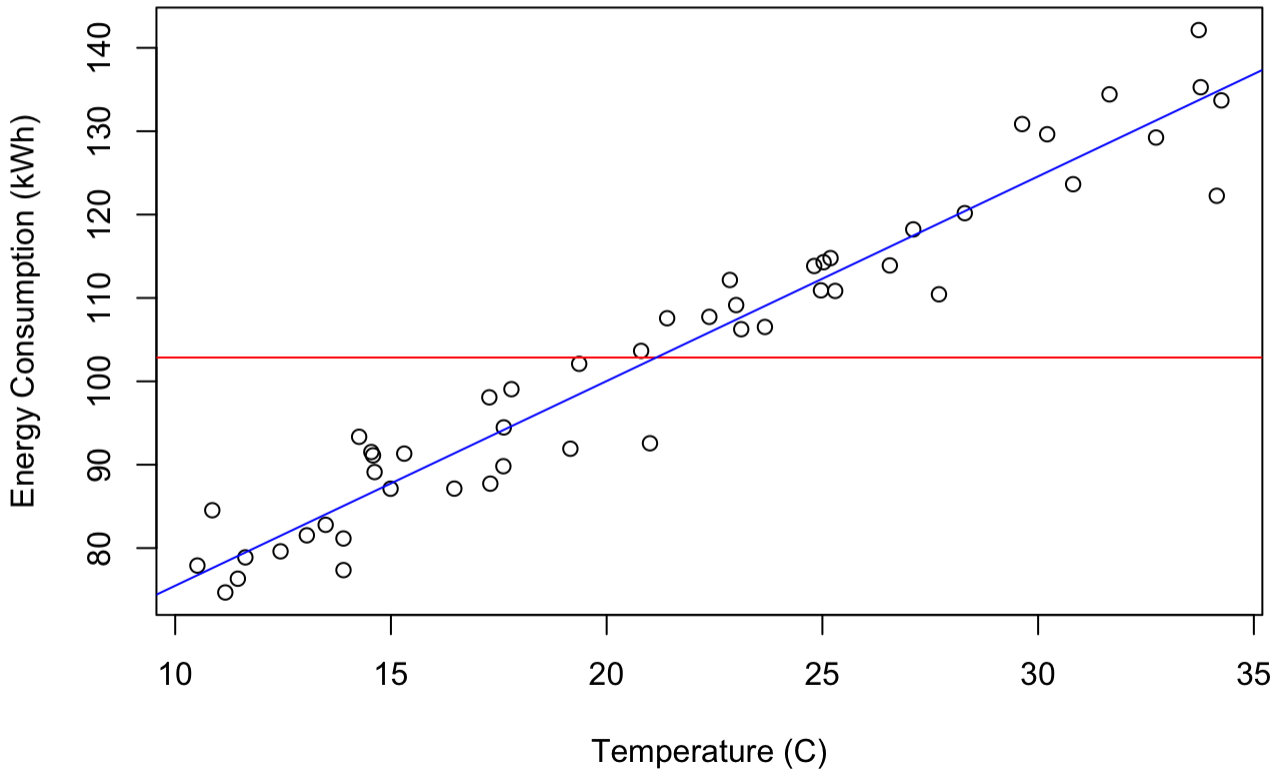
```
sse_lin
```

```
## [1] 1028.839
```

The simple linear regression model has a smaller SSE, therefore it is a better model.

```
plot(data$Temperature_C, data$Energy_Consumption_kWh, xlab = "Temperature (C)",  
      ylab = "Energy Consumption (kWh)", main = "Temperature vs Energy Consumption")  
abline(null_model, col = "red")  
abline(sim_lin_reg, col = "blue")
```

Temperature vs Energy Consumption



```
summary(null_model)
```

```
##
## Call:
## lm(formula = Energy_Consumption_kWh ~ 1, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.178 -14.780   0.021  11.335  39.273
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   102.86      2.59    39.71  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.31 on 49 degrees of freedom
```

1a. Null Model summary description

The residuals section shows the distribution of the residuals with min at -28.178 and max at 39.273.

Coefficients section:

- The value for $\hat{\beta}_0$ is 102.86, which is the mean of the data in the null model

- The standard error is 2.59, indicating that estimates for $\hat{\beta}_0$ deviate from the population parameter by 2.59 on average.
- The t-value is 39.71 meaning the value for $\hat{\beta}_0$ is 39.71 standard errors away from 0, and it's corresponding p-value (with a null hypothesis of $\beta_0 = 0$) is very small. This indicates that there is statistically significant evidence that $\beta_0 \neq 0$.
- A residual standard error of 18.31 indicates that the actual values deviate from their predicted values by 18.31 on average.

```
summary(sim_lin_reg)
```

```
##
## Call:
## lm(formula = data$Energy_Consumption_kWh ~ data$Temperature_C,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5034  -2.4541   0.0241   3.4235   8.3998
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    50.93031     2.04439   24.91  <2e-16 ***
## data$Temperature_C 2.45531     0.09158   26.81  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.63 on 48 degrees of freedom
## Multiple R-squared:  0.9374, Adjusted R-squared:  0.9361
## F-statistic: 718.8 on 1 and 48 DF,  p-value: < 2.2e-16
```

1b. Simple Linear Regression Model summary description

The residuals section shows the distribution of the residuals, with a min of -12.5034 and a max of 8.3998

Coefficients section:

- Intercept: $\hat{\beta}_0 = 50.93031$. It's standard error is 2.04439 meaning estimates for the intercept deviate from the true population parameter by 2.04439. It's t-value is 24.91 meaning $\hat{\beta}_0$ is 24.91 standard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_0 \neq 0$.
- `data$Temperature_C`: $\hat{\beta}_1 = 2.45531$. It's standard error is 0.09158 meaning estimates for the slope deviate from the true population parameter by 0.09158 on average. It's t-value is 26.81 meaning $\hat{\beta}_1$ is 26.81 standard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_1 \neq 0$.
- The residual standard error is 4.63, indicating the average deviation of 4.63 of actual values from their predicted values

- $R^2 = 0.9374$ indicating that around 93.74% of the variance in energy consumption is explained by temperature
- A large F-statistic and low p-value indicate that this model is better than just using the mean.

2. 95% confidence intervals and prediction intervals.

Null-model confidence interval for all estimates

```
ci = predict(null_model, interval="confidence", level = 0.95)
ci
```

```
##           fit      lwr      upr
## 1  102.8555  97.65047 108.0606
## 2  102.8555  97.65047 108.0606
## 3  102.8555  97.65047 108.0606
## 4  102.8555  97.65047 108.0606
## 5  102.8555  97.65047 108.0606
## 6  102.8555  97.65047 108.0606
## 7  102.8555  97.65047 108.0606
## 8  102.8555  97.65047 108.0606
## 9  102.8555  97.65047 108.0606
## 10 102.8555  97.65047 108.0606
## 11 102.8555  97.65047 108.0606
## 12 102.8555  97.65047 108.0606
## 13 102.8555  97.65047 108.0606
## 14 102.8555  97.65047 108.0606
## 15 102.8555  97.65047 108.0606
## 16 102.8555  97.65047 108.0606
## 17 102.8555  97.65047 108.0606
## 18 102.8555  97.65047 108.0606
## 19 102.8555  97.65047 108.0606
## 20 102.8555  97.65047 108.0606
## 21 102.8555  97.65047 108.0606
## 22 102.8555  97.65047 108.0606
## 23 102.8555  97.65047 108.0606
## 24 102.8555  97.65047 108.0606
## 25 102.8555  97.65047 108.0606
## 26 102.8555  97.65047 108.0606
## 27 102.8555  97.65047 108.0606
## 28 102.8555  97.65047 108.0606
## 29 102.8555  97.65047 108.0606
## 30 102.8555  97.65047 108.0606
## 31 102.8555  97.65047 108.0606
## 32 102.8555  97.65047 108.0606
## 33 102.8555  97.65047 108.0606
## 34 102.8555  97.65047 108.0606
## 35 102.8555  97.65047 108.0606
## 36 102.8555  97.65047 108.0606
## 37 102.8555  97.65047 108.0606
## 38 102.8555  97.65047 108.0606
## 39 102.8555  97.65047 108.0606
```

```
## 40 102.8555 97.65047 108.0606
## 41 102.8555 97.65047 108.0606
## 42 102.8555 97.65047 108.0606
## 43 102.8555 97.65047 108.0606
## 44 102.8555 97.65047 108.0606
## 45 102.8555 97.65047 108.0606
## 46 102.8555 97.65047 108.0606
## 47 102.8555 97.65047 108.0606
## 48 102.8555 97.65047 108.0606
## 49 102.8555 97.65047 108.0606
## 50 102.8555 97.65047 108.0606
```

Null-model prediction interval

```
pi = predict(null_model, interval="prediction", level = 0.95)
```

```
## Warning in predict.lm(null_model, interval = "prediction", level = 0.95): prediction
s on current data refer to _future_ responses
```

```
pi
```

```
##      fit      lwr      upr
## 1 102.8555 65.68403 140.027
## 2 102.8555 65.68403 140.027
## 3 102.8555 65.68403 140.027
## 4 102.8555 65.68403 140.027
## 5 102.8555 65.68403 140.027
## 6 102.8555 65.68403 140.027
## 7 102.8555 65.68403 140.027
## 8 102.8555 65.68403 140.027
## 9 102.8555 65.68403 140.027
## 10 102.8555 65.68403 140.027
## 11 102.8555 65.68403 140.027
## 12 102.8555 65.68403 140.027
## 13 102.8555 65.68403 140.027
## 14 102.8555 65.68403 140.027
## 15 102.8555 65.68403 140.027
## 16 102.8555 65.68403 140.027
## 17 102.8555 65.68403 140.027
## 18 102.8555 65.68403 140.027
## 19 102.8555 65.68403 140.027
## 20 102.8555 65.68403 140.027
## 21 102.8555 65.68403 140.027
## 22 102.8555 65.68403 140.027
## 23 102.8555 65.68403 140.027
## 24 102.8555 65.68403 140.027
## 25 102.8555 65.68403 140.027
## 26 102.8555 65.68403 140.027
## 27 102.8555 65.68403 140.027
## 28 102.8555 65.68403 140.027
## 29 102.8555 65.68403 140.027
```

```
## 30 102.8555 65.68403 140.027
## 31 102.8555 65.68403 140.027
## 32 102.8555 65.68403 140.027
## 33 102.8555 65.68403 140.027
## 34 102.8555 65.68403 140.027
## 35 102.8555 65.68403 140.027
## 36 102.8555 65.68403 140.027
## 37 102.8555 65.68403 140.027
## 38 102.8555 65.68403 140.027
## 39 102.8555 65.68403 140.027
## 40 102.8555 65.68403 140.027
## 41 102.8555 65.68403 140.027
## 42 102.8555 65.68403 140.027
## 43 102.8555 65.68403 140.027
## 44 102.8555 65.68403 140.027
## 45 102.8555 65.68403 140.027
## 46 102.8555 65.68403 140.027
## 47 102.8555 65.68403 140.027
## 48 102.8555 65.68403 140.027
## 49 102.8555 65.68403 140.027
## 50 102.8555 65.68403 140.027
```

Simple linear regression model confidence interval

```
ci_slr = predict(sim_lin_reg, interval="confidence", level=0.95)
ci_slr
```

##	fit	lwr	upr
## 1	98.47378	97.11695	99.83061
## 2	133.84098	131.17030	136.51166
## 3	120.41531	118.55328	122.27733
## 4	112.23080	110.73837	113.72322
## 5	85.06031	83.18576	86.93486
## 6	85.05883	83.18420	86.93346
## 7	79.04878	76.83058	81.26699
## 8	128.65179	126.31183	130.99175
## 9	112.38159	110.88380	113.87937
## 10	118.94695	117.16110	120.73279
## 11	76.74698	74.38762	79.10634
## 12	135.01926	132.27135	137.76716
## 13	126.58113	124.36782	128.79444
## 14	88.51742	86.81766	90.21718
## 15	86.64438	84.85246	88.43630
## 16	86.74133	84.95434	88.52833
## 17	94.15870	92.68955	95.62784
## 18	107.69448	106.32894	109.06002
## 19	101.99745	100.67944	103.31547
## 20	93.35992	91.86322	94.85661
## 21	113.04071	111.51873	114.56269
## 22	84.04597	82.11653	85.97542
## 23	93.41611	91.92141	94.91081
## 24	97.97177	96.60534	99.33821

```
## 25 103.47831 102.16104 104.79558
## 26 123.67977 121.63727 125.72227
## 27 87.73999 86.00276 89.47722
## 28 107.04861 105.69514 108.40209
## 29 111.84753 110.36843 113.32663
## 30 78.33470 76.07318 80.59623
## 31 112.77627 111.26414 114.28840
## 32 85.95070 84.12308 87.77831
## 33 79.47650 77.28403 81.66897
## 34 133.72873 131.06537 136.39208
## 35 134.75667 132.02604 137.48731
## 36 125.10516 122.97981 127.23051
## 37 94.18150 92.71312 95.64989
## 38 81.47884 79.40449 83.55319
## 39 117.48361 115.77001 119.19721
## 40 102.50125 101.18454 103.81796
## 41 82.97450 80.98556 84.96344
## 42 105.87881 104.54298 107.21463
## 43 77.59431 75.28741 79.90121
## 44 131.30011 128.79346 133.80675
## 45 91.36809 89.79483 92.94136
## 46 116.15094 114.49953 117.80235
## 47 94.61715 93.16295 96.07136
## 48 107.40669 106.04673 108.76666
## 49 109.04207 107.64627 110.43787
## 50 86.83034 85.04785 88.61283
```

Simple linear regression model prediction interval

```
pi_slr = predict(sim_lin_reg, interval="prediction", level=0.95)
```

```
## Warning in predict.lm(sim_lin_reg, interval = "prediction", level = 0.95): predictions on current data refer to _future_ responses
```

```
pi_slr
```

```
##           fit          lwr          upr
## 1  98.47378  89.06677 107.88079
## 2 133.84098 124.15680 143.52516
## 3 120.41531 110.92226 129.90835
## 4 112.23080 102.80328 121.65832
## 5  85.06031  75.56480  94.55582
## 6  85.05883  75.56331  94.55436
## 7  79.04878  69.47950  88.61807
## 8 128.65179 119.05355 138.25003
## 9 112.38159 102.95322 121.80996
## 10 118.94695 109.46855 128.42534
## 11  76.74698  67.14399  86.34997
## 12 135.01926 125.31350 144.72502
## 13 126.58113 117.01298 136.14928
## 14  88.51742  79.05487  97.97998
```



```
## 15 86.64438 77.16483 96.12392
## 16 86.74133 77.26272 96.21995
## 17 94.15870 84.73483 103.58256
## 18 107.69448 98.28621 117.10275
## 19 101.99745 92.59597 111.39894
## 20 93.35992 83.93172 102.78811
## 21 113.04071 103.60847 122.47295
## 22 84.04597 74.53947 93.55247
## 23 93.41611 83.98823 102.84399
## 24 97.97177 88.56338 107.38017
## 25 103.47831 94.07693 112.87969
## 26 123.67977 114.14968 133.20986
## 27 87.73999 78.27063 97.20935
## 28 107.04861 97.64209 116.45514
## 29 111.84753 102.42211 121.27295
## 30 78.33470 68.75528 87.91412
## 31 112.77627 103.34561 122.20693
## 32 85.95070 76.46434 95.43706
## 33 79.47650 69.91314 89.03985
## 34 133.72873 124.04656 143.41089
## 35 134.75667 125.05579 144.45756
## 36 125.10516 115.55697 134.65335
## 37 94.18150 84.75776 103.60524
## 38 81.47884 71.94187 91.01580
## 39 117.48361 108.01855 126.94866
## 40 102.50125 93.09995 111.90256
## 41 82.97450 73.45574 92.49325
## 42 105.87881 96.47481 115.28281
## 43 77.59431 68.00408 87.18454
## 44 131.30011 121.65987 140.94034
## 45 91.36809 81.92744 100.80875
## 46 116.15094 106.69695 125.60493
## 47 94.61715 85.19561 104.03870
## 48 107.40669 97.99923 116.81415
## 49 109.04207 99.62936 118.45478
## 50 86.83034 77.35257 96.30810
```

3. Which model

The simple linear regression model has a smaller RSE, higher R^2 value and higher F-value so therefore I would rather have the simple linear regression model over the null model.

4. Interpretations

4a. Null Model

- The intercept is 102.86, meaning that the energy consumed is 102.86 kWh on average.
- The standard error is 2.59, indicating that estimates for the intercept deviate from the population parameter by 2.59 on average.

- The t-value is 39.71 meaning the value for $\hat{\beta}_0$ is 39.71 standard errors away from 0, and it's corresponding p-value (with a null hypothesis of $\beta_0 = 0$) is very small. This indicates that there is statistically significant evidence that $\beta_0 \neq 0$.

4b. Simple Linear Regression Model

- Intercept: $\hat{\beta}_0 = 50.93031$. This means that when the temperature is 0 degrees Celsius, the energy consumed is 50.93031 kWh. It's standard error is 2.04439 meaning estimates for the intercept deviate from the true population parameter by 2.04439. It's t-value is 24.91 meaning $\hat{\beta}_0$ is 24.91 standard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_0 \neq 0$.
- `data$Temperature_C`: $\hat{\beta}_1 = 2.45531$. This means that for each 1 degree Celsius increase in temperature, the energy consumed increases by 2.45531 kWh. It's standard error is 0.09158 meaning estimates for the slope deviate from the true population parameter by 0.09158 on average. It's t-value is 26.81 meaning $\hat{\beta}_1$ is 26.81 standard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_1 \neq 0$.
- The residual standard error is 4.63, indicating the average deviation of 4.63 of actual values from their predicted values
- $R^2 = 0.9374$ indicating that around 93.74% of the variance in energy consumption is explained by temperature
- A large F-statistic and low p-value indicate that this model is better than just using the mean.